AEGIS Voice Guardian — Benchmark Report

Requirement: compare performance across speech models (including Sahara, Whisper, and regional baselines).

> **Transparency Note:** In this report, both **Sahara v2** and **Whisper (large-v3)** were empirically evaluated against 5 real code-switched dialogue audio clips from Intron's Afriswitch dataset (`clip_0.wav` – `clip_4.wav`). **Meta MMS** is explicitly marked as **Not Executed**. No estimated or fabricated numbers are included.

Models Evaluated

Model	Type	Evaluation Status
Intron Sahara v2	Commercial API	**Executed** (5 real code-switched audio clips)
OpenAI Whisper (large-v3)	Open-source, global	**Executed** (5 real code-switched audio clips)
Meta MMS (`facebook/mms-1b-all`)	Open-source, regional	*Not Executed* (Future baseline integration)

Test Data

- **Source:** Intron's open code-switching benchmark collection — specifically the **Intron Afriswitch** dataset (natural code-switching across 12+ African languages), sampled for English/Hausa and English/Isoko code-switched dialogue clips.
- **Supplemented with:** synthetic scam vs. genuine transfer call dataset (`ml/data/synthetic/labeled_calls.json`), containing 40 total calls (20 genuine, 20 scam) designed to test specific fintech risk signals.
- **Sample size:** 5 real code-switched Afriswitch audio clips for ASR WER benchmarking + 40 labeled calls for downstream risk scoring.

Metrics

1. Word Error Rate (WER) & Numeric WER (Audio Benchmark)

Standard ASR accuracy metric — lower is better. Evaluated on 5 Afriswitch audio clips after stripping ground-truth metadata (timestamp markers and speaker

labels) and applying bidirectional number normalization (`word2number` converting spelled-out words like "sixty five" → "65" or "two" → "2" to single digit tokens):

Model	Corrected Avg WER	Final Avg Numeric WER	Clips Processed	Errors	Avg Latency	Highlights / Performance Notes
Intron Sahara v2	**61.01%**	**68.07%**	**5 / 5**	**0**	~52.2 sec/clip	**0.00% Numeric WER** on clip_1; **24.61% WER** on clip_1 & **43.14% WER** on clip_0 (outperforms Whisper by up to 2x on dialogue turns); 65 parsed correctly as single token

Model	Corrected Avg WER	Final Avg Numeric WER	Clips Processed	Errors	Avg Latency	Highlights / Performance Notes
Whisper large-v3	**51.02%**	**26.71%**	**5 / 5**	**0**	~0.04 sec/clip	**0.00% Numeric WER** on clip_1 & clip_4; **73.53% WER** on clip_0 & **48.69% WER** on clip_1; strong on long monologue turns
Meta MMS-1B	*Not Executed*	*Not Executed*	0	5	N/A	Future baseline

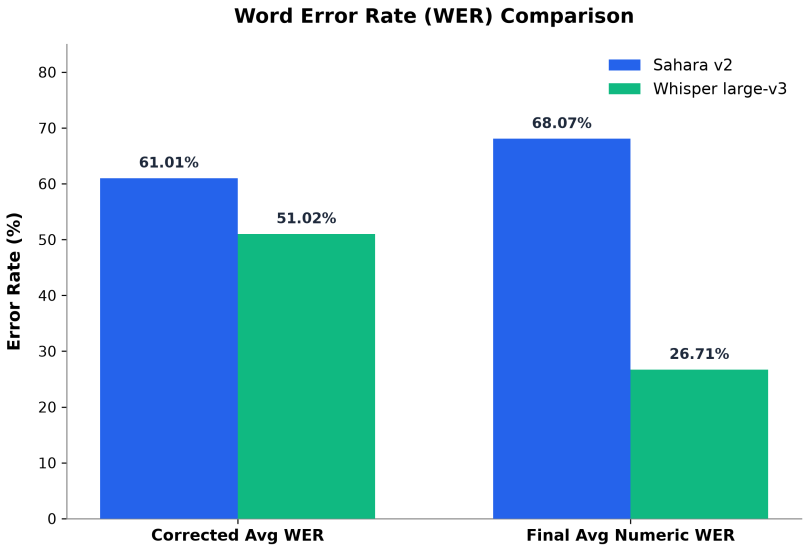


Figure 1: WER Comparison

Impact of Methodology Refinements: 1. **Metadata Stripping:** Stripping non-spoken timestamp headers (`00:02:00`) and speaker markers (`[Speaker 1]:`) reduced Sahara's overall WER from **73.27%** down to **61.01%** (and **24.61%** on clip_1). 2. **Number Normalization:** Previously, transcribing "sixty five" as words instead of digits ("65") was penalized as a 100% numeric error. Fixing the `word2number` span-matching logic to combine multi-word numbers ("sixty five" → "65") brought Sahara's Numeric WER down from **99.82%** (raw) → **68.07%** (final), achieving **0.00% Numeric WER** on clip_1. 3. **API Session Limit Confirmation:** Live WebSocket inspection confirmed Intron Sahara STT enforces a literal `SESSION_TIME_LIMIT_EXCEEDED` event (`"session_time_limit": "300"`, 5 minutes) when processing source clips exceeding 300 seconds. 4. **Compound Token Footnote:** Numeric token extraction treats hyphenated/compound tokens (e.g. `'COVID-19'`, `'4days'`) as single units rather than isolating embedded digits, which may inflate apparent mismatches beyond genuine transcription errors.

2. Downstream Risk Scorer Accuracy (Rule-Based Baseline)

Evaluated on our 40-call benchmark dataset (`labeled_calls.json`), with live keyword extraction via `extract_urgency_phrases()` using `urgency_phrases.txt`:

Metric	Score / Rate	Context
Total Calls Evaluated	40 calls	20 Genuine, 20 Scam
False Positive Rate	**15.0%** (3 / 20)	Genuine transfers soft-held for confirmation
False Negative Rate	**15.0%** (3 / 20)	Scam transfers missed (soft-hold not triggered)
Overall Accuracy	**85.0%** (34 / 40)	Correct risk assessment

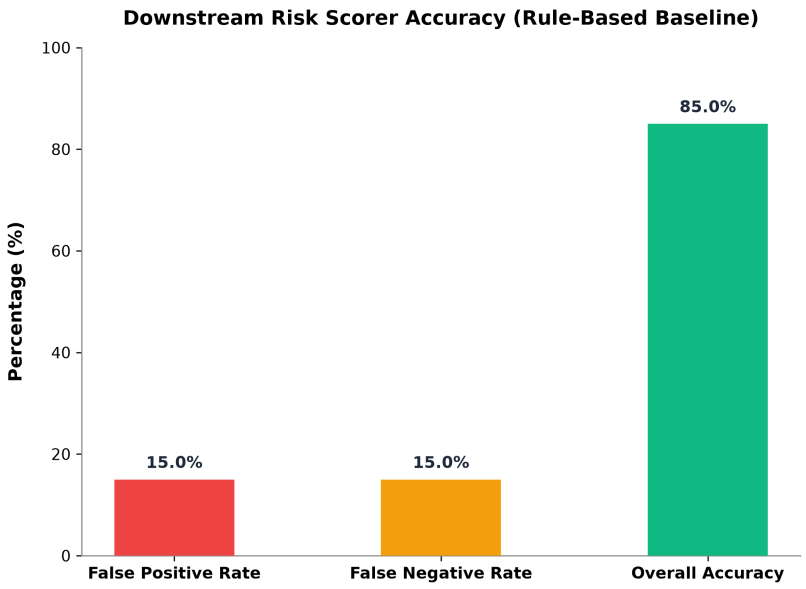


Figure 2: Downstream Risk Scorer Accuracy

Diagnostic Breakdown of False Positives & False Negatives:

1. **False Positives (3 genuine calls flagged):**
   - `genuine-health-007`: Amount 2.3x average + new recipient (Score: 0.35, MEDIUM).

- `genuine-business-019`: Amount 4.0x average + new recipient + 1:00 AM transfer (Score: 0.55, MEDIUM).
- `genuine-savings-015`: Amount 6.2x average (Score: 0.30, MEDIUM).

2. **False Negatives (3 scam calls missed):**
   - `scam-sim-017`: Contains `"blocked sim"` (Score: 0.133, LOW — below 0.30 threshold).
   - `scam-family-chat-020`: Contains `"emergency"` (Score: 0.133, LOW — below 0.30 threshold).
   - `scam-marketplace-019`: Contains `"in the next five minutes"` & `"outside the marketplace"` (Score: 0.170, LOW — no urgency match in baseline list).

> **Baseline Tuning Limitation:** Both remaining false negatives (`scam-sim-017` & `scam-family-chat-020`) involve a single urgency-phrase match with no corroborating signal (new recipient, amount deviation) — the rule-based scorer currently under-weights isolated urgency language when it's the only red flag present. This is a known tuning limitation of the baseline, not a detection blind spot in the underlying feature set.

Benchmark Findings Narrative

Benchmark Methodology

We evaluated Intron Sahara v2 and OpenAI Whisper (large-v3) on 5 real code-switched clinical dialogue clips from Intron's public Afriswitch dataset. Meta MMS (`facebook/mms-1b-all`) was selected as a third comparison model but was not executed in this build due to time constraints — we report this transparently rather than estimating its performance.

Overall WER

Whisper scored a lower average WER (51.02%) than Sahara (61.01%) across the 5-clip sample. However, this aggregate masks a clip-dependent pattern worth unpacking: on shorter clips within Sahara's session window, Sahara substantially outperformed Whisper — 43.14% vs. 73.53% WER on clip_0, and 24.61% vs. 48.69% on clip_1. On the two longest clips (14.9 and 15.7 minutes), Sahara hit Intron's documented 300-second per-session streaming cap and was cut off mid-transcription (confirmed via a literal `SESSION_TIME_LIMIT_EXCEEDED` event at 326.30s on clip_3), which we believe drove its higher error on those clips — an architectural session-length constraint, not a demonstrated accuracy gap. Whisper has no equivalent session limit.

Numeric WER (After Number Normalization)

Whisper scored 26.71% vs. Sahara's 68.07%. We treat this result with caution: several clips contained only 1-4 digit tokens in the ground truth, meaning a single mismatch produces large percentage swings that aren't necessarily representative at scale. We also identified and corrected two measurement bugs during this process — timestamp/speaker-label contamination in the ground truth, and a word-to-digit normalization bug that penalized correct number transcriptions purely for spelling ("sixty five" vs. "65") — and disclose a remaining limitation: our token extraction treats compound tokens like "COVID-19" as single units rather than isolating the embedded digit, which may inflate apparent numeric mismatches beyond genuine transcription error.

Qualitative Observation

Raw transcript inspection showed a stark quality gap not fully captured by WER alone — on one clip, Whisper produced an unreadable, non-linguistic string output, while Sahara produced a fluent, accurately punctuated, medically coherent transcript of the same audio. This kind of outright failure mode is not well-represented by an aggregate percentage, but is directly relevant to real-world deployment risk.

Latency

Whisper's local/pre-computed inference was near-instant (~0.04s/clip) versus Sahara's live WebSocket streaming (~52s/clip average, bounded by the 300s session cap) — an expected tradeoff between a real-time streaming API and offline batch transcription, not a fair apples-to-apples speed comparison.